

SQL Code

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1  CREATE TABLE departments (  
2  dept_id INT PRIMARY KEY,  
3  dept_name VARCHAR(50),  
4  location VARCHAR(50)  
5  );  
6  INSERT INTO departments VALUES  
7  (10,'HR','Jaipur'),  
8  (20,'IT','Bangalore'),  
9  (30,'Finance','Mumbai'),  
10 (40,'Marketing','Delhi'),  
11 (50,'Legal','Pune');  
12  
13 CREATE TABLE employeeess (  
14 emp_id INT PRIMARY KEY,  
15 emp_name VARCHAR(50),  
16 dept_id INT,  
17 salary INT,  
18 city VARCHAR(50)  
19 );  
20 INSERT INTO employeeess VALUES  
21 (101,'Amit',10,45000,'Jaipur'),  
22 (102,'Bhavna',20,72000,'Bangalore'),  
23 (103,'Chirag',20,68000,'Bangalore'),  
24 (104,'Divya',30,55000,'Mumbai'),  
25 (105,'Esha',NULL,39000,'Delhi'),  
26 (106,'Farhan',40,61000,'Delhi'),  
27 (107,'Gaurav',60,50000,'Kolkata'),  
28 (108,'Hina',30,83000,'Mumbai');  
29 CREATE TABLE projects (  
30 proj_id INT PRIMARY KEY,  
31 proj_name VARCHAR(50),  
32 dept_id INT,  
33 budget INT  
34 );INSERT INTO projects VALUES  
35 (1,'Payroll Revamp',10,120000),  
36 (2,'Cloud Migration',20,500000),  
37 (3,'Mobile App',20,300000),  
38 (4,'Audit Automation',30,250000),  
39 (5,'Brand Refresh',40,180000),  
40 (6,'Data Lake',70,400000);  
41  
42 -- Q1. List each employee with their department name and location.  
43 select e.emp_id, e.emp_name, d.dept_name, d.location from employeeess as e  
44 inner join departments as d  
45 on e.dept_id=d.dept_id;  
46  
47 -- Q2. Show every project along with the name of the department that owns it  
48 select p.proj_id, p.proj_name, d.dept_name ,p.budget  
49 from projects as p  
50 join departments as d
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51 on p.dept_id=d.dept_id;
52
53 -- Q3. Show employee names together with the projects run by their own department
54 select e.emp_id, e.emp_name , d.dept_id , d.dept_name
55 from employees as e
56 join departments as d
57 on e.dept_id=d.dept_id;
58
59 -- Q4. List ALL employees with their department name; show NULL when the employee has
    no matching departments
60 select e.emp_id, e.emp_name, e.dept_name, d.dept_id
61 from employees as e
62 left join departments as d
63 on e.dept_id=d.dept_id;
64
65 -- Q5. List ALL departments with their projects; departments with no project must
    still appear
66 select d.dept_id, d.dept_name, p.proj_name, p.budget
67 from departments as d
68 left join projects as p
69 on p.dept_id=d.dept_id;
70
71 -- Q6. Find only those employees who do NOT belong to any valid department.
72 select e.emp_id, e.emp_name, e.dept_id
73 from employees as e
74 left join departments as d
75 on e.dept_id=d.dept_id
76 where d.dept_id is null;
77
78 -- Q7. List ALL departments and any employees in them, using a RIGHT JOIN with
    employees on the left.
79 select e.emp_name, d.dept_id, d.dept_name
80 from employees as e
81 right join departments as d
82 on e.dept_id=d.dept_id;
83
84 -- Q8. List ALL projects and the department that owns them, keeping projects that
    point to a missing department.
85 select d.dept_name , p.proj_id, p.proj_name , p.dept_id
86 from departments as d
87 right join projects as p
88 on d.dept_id=p.dept_id;
89
90 -- Q9. Find departments that currently have NO employee assigned.
91 select d.dept_id , d.dept_name, d.location
92 from employees as e
93 right join departments as d
94 on e.dept_id=d.dept_id where e.emp_id is null;
95
96 -- Q10. Produce every possible pairing of the Finance/Legal departments with employees
    earning above 70000.
97 select e.emp_name, e.salary, d.dept_name
98 from employees e
99 cross join departments d
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100 where e.salary > 70000
101 and d.dept_name in ('Finance', 'Legal');
102
103 -- Q11. Pair every project having a budget of at least 400000 with every department
    located in Mumbai or Pune.
104 select p.proj_name, p.budget, d.dept_name, d.location
105 from projects p
106 cross join departments d
107 where p.budget >= 400000
108 and d.location in ('Mumbai', 'Pune');
109
110 -- Q12. How many total row combinations result from CROSS JOIN of employees and
    departments? Show the count
111 select count(*) as total_combinations
112 from employeeess
113 cross join departments;
114
115 -- Q13. Show employees whose salary is greater than 60000
116 select *
117 from employeeess
118 where salary > 60000;
119
120 -- Q14. Show all employees based in Delhi or Mumbai.
121 select emp_id, emp_name, salary, city
122 from employeeess
123 where city in ('Delhi', 'Mumbai');
124
125 -- Q15. Show employees whose dept_id is NULL (unassigned employees)
126 select emp_id, emp_name, dept_id, salary
127 from employeeess
128 where dept_id is null;
129
130 -- Q16. Show departments located in Bangalore or Delhi
131 select *
132 from departments
133 where location in ('Bangalore', 'Delhi');
134
135 -- Q17. Show departments whose dept_id is 30 or higher.
136 select *
137 from departments
138 where dept_id >= 30;
139
140 -- Q18. Show departments whose name starts with the letter "F" or "L".
141 select *
142 from departments
143 where dept_name like 'F%'
144 or dept_name like 'L%';
145
146 -- Q19. Show projects with a budget between 200000 and 400000 (inclusive).
147 select *
148 from projects
149 where budget between 200000 and 400000;
150
151 -- Q20. Show all projects that belong to dept_id 20.
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152 select *
153 from projects
154 where dept_id = 20;
155
156 -- Q21. Show projects whose name contains the word "a" and budget is under 300000.
157 select proj_id, proj_name, budget
158 from projects
159 where proj_name like '%a%'
160 and budget < 300000;
161
162 -- Q22. Show employee name, department name, location and project name for all
    matching rows across all three
163 select e.emp_name, d.dept_name, d.location, p.proj_name, p.budget
164 from employeeess e
165 join departments d
166 on e.dept_id = d.dept_id
167 join projects p
168 on d.dept_id = p.dept_id;
169
170 -- Q23. Show ALL employees, plus department and project details where available (keep
    employees even with no dept/project)
171 select e.emp_id, e.emp_name, d.dept_name, p.proj_name
172 from employeeess e
173 left join departments d
174 on e.dept_id = d.dept_id
175 left join projects p
176 on d.dept_id = p.dept_id;
177
178 -- Q24. Show employees earning more than 60000 along with department and any project
    over 250000 budget.
179 select e.emp_name, e.salary, d.dept_name, p.proj_name, p.budget
180 from employeeess e
181 join departments d
182 on e.dept_id = d.dept_id
183 join projects p
184 on d.dept_id = p.dept_id
185 where e.salary > 60000
186 and p.budget > 250000;
187
188 -- Q25. List ALL departments with their employees and projects, including departments
    having neither.
189 select d.dept_id, d.dept_name, e.emp_name, p.proj_name
190 from employeeess e
191 right join departments d
192 on e.dept_id = d.dept_id
193 left join projects p
194 on d.dept_id = p.dept_id;
195
196 -- Q26. Show employees who work in a department located in Bangalore or Mumbai, along
    with the projects of that department.
197 select e.emp_name, d.location, p.proj_name
198 from employeeess e
199 join departments d
200 on e.dept_id = d.dept_id
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201 join projects p
202 on d.dept_id = p.dept_id
203 where d.location in ('Bangalore', 'Mumbai');
204
205 -- Q27. Pair employees who work in the same city (avoid duplicate pairs and self-
pairing)
206 select e1.emp_name as employee_1,
207        e2.emp_name as employee_2,
208        e1.city
209 from employees e1
210 join employees e2
211 on e1.city = e2.city
212 where e1.emp_id < e2.emp_id;
213
214 -- Q28. Show every project along with employees of that department; keep projects with
no employees.
215 select p.proj_name, p.dept_id, e.emp_name
216 from projects p
217 left join employees e
218 on p.dept_id = e.dept_id;
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